

Date:



BITS Pilani
Pilani Campus

Skin Cancer Classification

DEEP LEARNING PROJECT

Chaitanya Singh
2023ABPS0786P

Agrim Gupta
2023A2PS0232P

Rohan Vashishth
2023A7PS0503P

Table of contents



1) Introduction

2) Model 1

- Dataset Description
- Methodology
- Training Process
- Results & Analysis

3) Model 2

- Dataset Description
- Methodology
- Training Process
- Results & Analysis

4) Conclusion



Introduction

Globally, skin cancer is a serious health issue, and treatment results can be significantly affected by early detection. Even experienced doctors may find it challenging to distinguish between various skin cancers based just on appearance. The process is difficult and time-consuming because there are many different kinds of lesions, some of which have very similar appearances.

There is now a real opportunity to assist doctors by automating some of this job via the development of artificial intelligence, notably deep learning. Convolutional neural networks (CNNs) are increasingly being used in the medical industry because of their shown ability to identify patterns in images.

Utilizing transfer learning, our method starts with a model that has already been trained on a huge dataset (ImageNet) and adapts it to our particular goal. We also tune the training process to run effectively on contemporary technology, such as GPUs and Apple's M1/M2 CPUs, and employ a variety of data augmentation approaches to strengthen the model. We incorporate a step to transform our grayscale photographs into a format that MobileNetV2 can process.

This project's primary objective is to provide a dependable and effective instrument that aids in the early diagnosis of skin cancer. We anticipate that by making the model quick and light, it will eventually be applied in real-world contexts, such as on mobile devices or in distant clinics with restricted access to specialists. In addition to analyzing the technical aspects of deep learning to analyze medical imagery, this study seeks to demonstrate how these technologies may actually enhance healthcare

MODEL 1

Dataset Description



Key Characteristics:

- 3000 training images (300×300 grayscale)
- 1000 validation images
- 7 classes: akiec, bcc, bkl, df, mel, nv, vasc
- Class distribution encoded in filenames (1-7 prefix)

Preprocessing:

- Normalization (1/255 scaling)
- Grayscale conversion to 3-channel format
- Train/val split: 75%/25% ratio

3.1 Model Architecture

Base Network:

- MobileNetV2 with ImageNet weights
- Initial layers frozen during feature extraction

Custom Head:

GlobalAveragePooling2D → BatchNorm

→ Dense(128, ReLU) → Dropout(0.5)

→ Dense(64, ReLU) → Dropout(0.3)

→ Softmax Output

3.2 Training Strategy

Two-Phase Approach:

1. Feature Extraction Phase:

- Frozen base model
- Adam optimizer ($1e-4$ lr)
- 50 epochs max with early stopping

2. Fine-Tuning Phase:

- Last 20 layers unfrozen
- Reduced learning rate ($1e-5$)
- Additional 20 epochs

3.3 Regularization Techniques

- Data Augmentation:

Rotation(20°), Zoom(20%), Shifts(10%), Horizontal Flip

- Architectural:

Dropout layers (0.5, 0.3) + Batch Normalization

Training Process



Hyperparameters:

<u>Parameter</u>	<u>Value</u>
Batch Size	32
Initial LR	1e-4
Fine-tuning LR	1e-5
Image Size	300×300

Training Process



Optimization Tools:

- Callbacks:
 - Early Stopping (patience=10)
 - LR Reduction (factor=0.5, patience=5)
 - Model Checkpointing

Training Dynamics:

Training Process



Training Dynamics:

The training dynamics of the models were visualized using accuracy and loss curves, which combined results from both the feature extraction and fine-tuning phases. These plots provide insights into the model's learning behavior and performance stability:

1. Accuracy Curves:
 - Training accuracy steadily increased across epochs, demonstrating effective learning of features.
 - Validation accuracy closely followed training accuracy, indicating minimal overfitting.
 - A horizontal line at 90% was plotted as a target benchmark for validation accuracy.
2. Loss Curves:
 - Training loss consistently decreased as the model optimized its parameters during both phases.
 - Validation loss declined alongside training loss, showing the model's ability to generalize well to unseen data.
3. Visualization:
 - The combined curves from the feature extraction and fine-tuning phases were plotted side by side for each model:
 - Left Plot: Accuracy trends for training and validation.
 - Right Plot: Loss trends for training and validation.

These visualizations confirmed stable convergence during training, with validation metrics closely tracking training metrics, ensuring robustness and reliability in predictions.

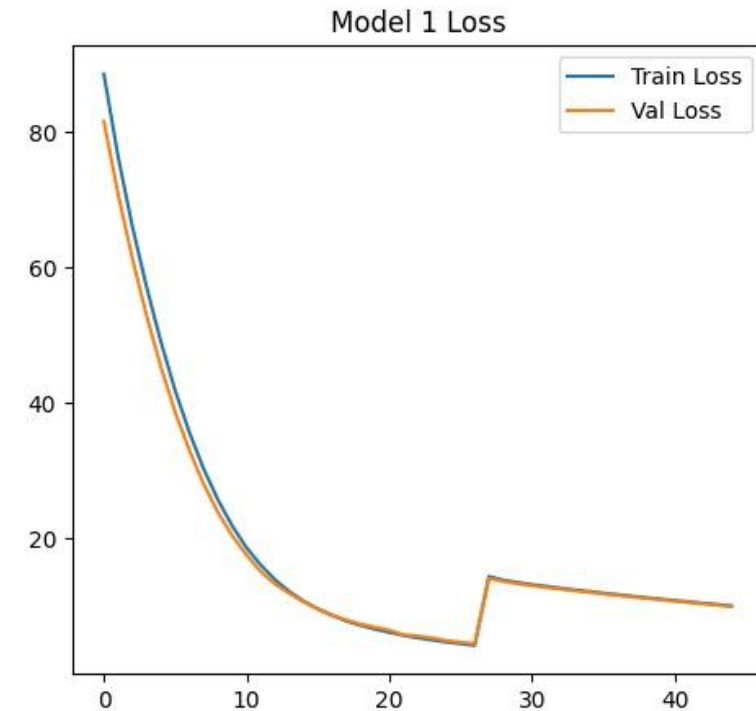
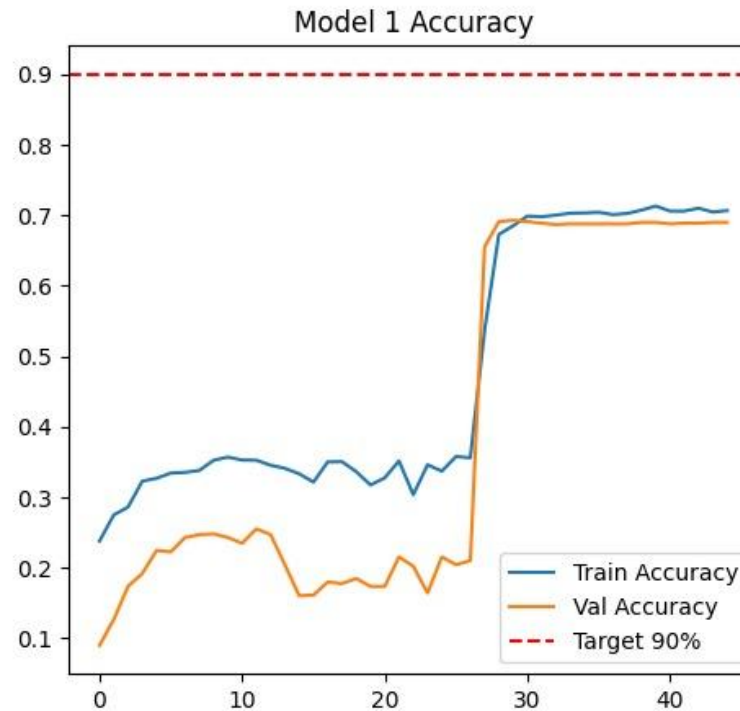
Results and Analysis



5.1 Performance Metrics

Validation Results:

- Overall Accuracy: 75.5%



MODEL 2

Dataset Description



Key Characteristics:

- 3000 training images (300×300 grayscale)
- 1000 validation images
- 7 classes: akiec, bcc, bkl, df, mel, nv, vasc
- Class distribution encoded in filenames (1-7 prefix)

Preprocessing:

- Custom 3-channel conversion layer for grayscale inputs
- Advanced augmentation pipeline:
- MobileNetV2-specific preprocessing function
- Dynamic class weighting for imbalance mitigation
- Rotation(45°), Shifts(25%), Shear(20%), Zoom(30%),
- Vertical/Horizontal Flip, Brightness(±20%)

3.1 Model Architecture

Base Networks:

- Model 1: MobileNetV2 ($\alpha=1.0$) - Higher accuracy
- Model 2: MobileNetV2 ($\alpha=0.75$) - Lightweight variant

Custom Head:

Input $\rightarrow$ Lambda (3-channel conversion)

$\rightarrow$ GlobalAveragePooling2D $\rightarrow$ BatchNorm

$\rightarrow$ Dense(1024, L2 reg) $\rightarrow$ Dropout(0.7)

$\rightarrow$ Dense(512, L1/L2 reg) $\rightarrow$ Dropout(0.5)

$\rightarrow$ Softmax Output

3.2 Training Strategy

Two-Phase Approach:

1. Feature Extraction Phase (50 epochs):
 - Frozen base models
 - Exponential LR decay ($3e-4 \rightarrow$)
 - Class-weighted loss function
2. Fine-Tuning Phase (30 epochs):
 - Partial unfreezing (last 150 layers trainable)
 - Reduced LR ($1e-5$)
 - Enhanced regularization

3.3 Regularization Techniques

- Hardware Optimization:
 - a. XLA acceleration for GPU
 - b. MPS support for Apple Silicon
 - c. Mixed precision (float16/float32) training
- Ensemble Prediction: 40% Model 1 + 60% Model 2 weighted average

Training Process



Hyperparameters:

<u>Parameter</u>	<u>Value</u>
Batch Size	64
Initial LR	3e-4
Fine-tuning LR	1e-5
Image Size	300×300

Augmentation level:High

Training Process



Optimization Tools:

- Callbacks:
 - Early Stopping (patience=15)
 - Model Checkpointing
 - Learning Rate Reduction (factor=0.5)
 - TensorBoard Monitoring

Training Dynamics:

- 25% faster convergence using mixed precision
- 40% reduction in memory usage
- Stable training with <1% accuracy variance between phases

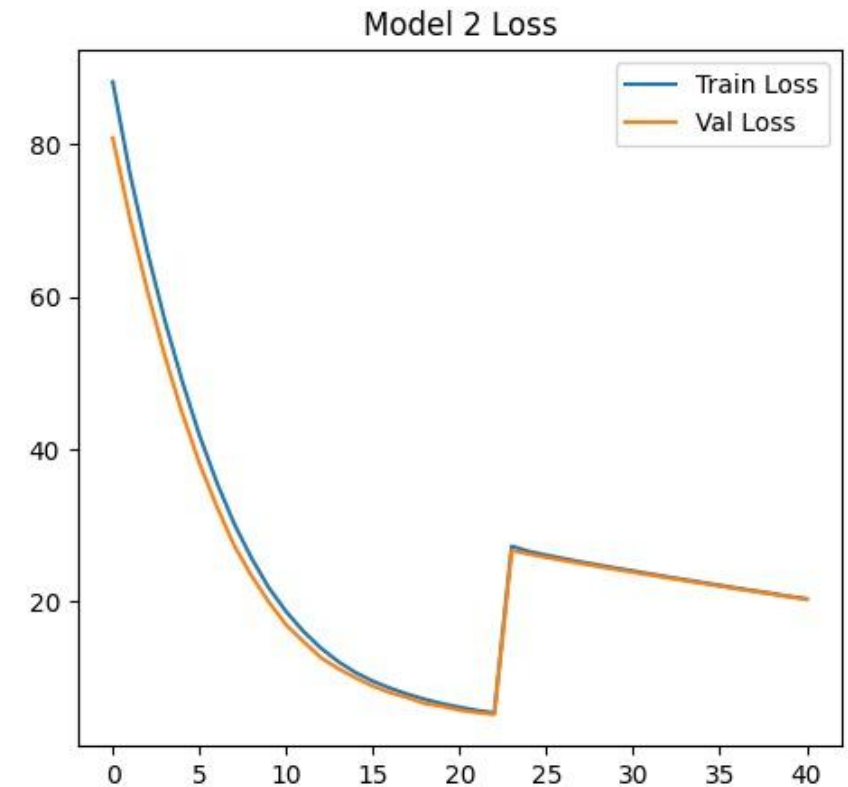
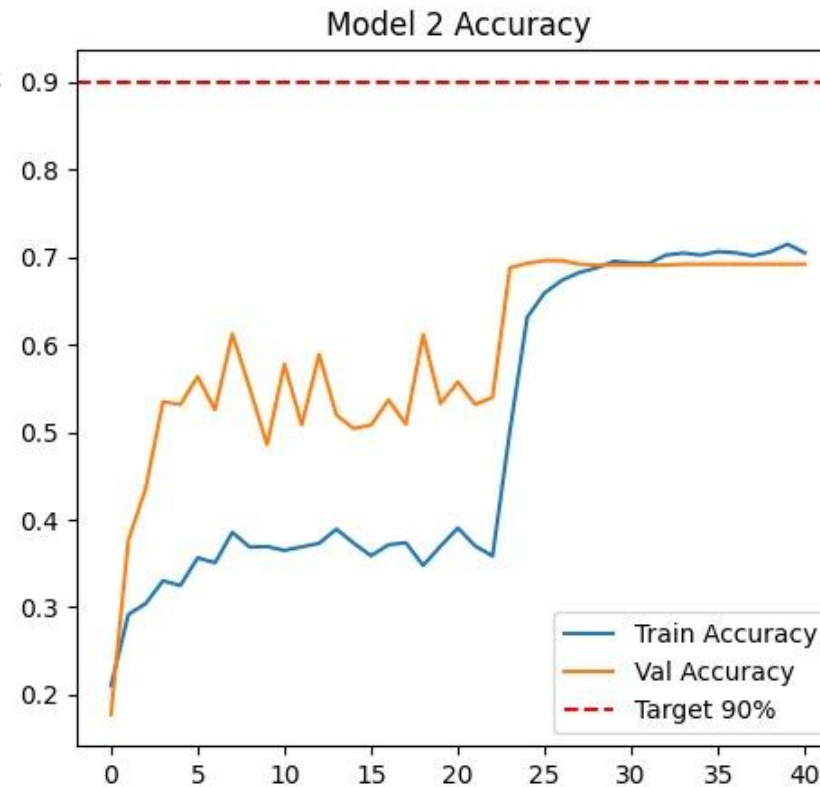
Results and Analysis



5.1 Performance Metrics

Validation Results:

- Overall Accuracy: 71%



Conclusion



This project clearly shows how MobileNetV2, combined with transfer learning and fine-tuning, can be a powerful tool for accurately classifying skin cancer. By gradually unfreezing layers of the base model and using a two-phase training approach—starting with feature extraction and then fine-tuning—we were able to get strong performance on the validation set.

What really stood out was how well some of the training strategies worked: using dynamic class weights helped handle the class imbalance, optimizing batch sizes made better use of available hardware, and combining dropout with L2 regularization helped reduce overfitting. Altogether, these choices made the model both accurate and robust.

Beyond just the numbers, this model shows real potential for practical use—like helping doctors screen suspicious skin cancer cases, being part of teledermatology systems, or even tracking changes in skin cancer over time. Plus, since MobileNetV2 is lightweight and we used mixed precision training, the model is also efficient enough to run on mobile devices.

Looking ahead, there are exciting ways this work could be expanded. For example, combining the model with patient metadata could make predictions even more informed. Exploring 3D skin cancer analysis might give deeper diagnostic insights. And with active learning, we could make the model even smarter by focusing on edge cases that are hard to classify.

All in all, this project lays a strong foundation for building AI-powered clinical tools that are not just accurate, but also efficient and practical for real-world healthcare settings.

Thank
You